RELATIVE PERFECTNESS OVER A FILTERED COLIMIT OF RINGS DESCENDS TO A STAGE, AND HOM GROUPS COMMUTE WITH THE COLIMIT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed. The counted statement is the labeled lemma environment at source lines 22104-22127 (canonical tag 0DHX) and the counted proof is the proof environment at source lines 22129-22161, which immediately follows it and is its own proof.

Editorial note. The printed text writes that category of R-flat, finitely presented A-modules without the article and begins the last sentence with Thus the result because; both are copied exactly as printed and are not corrected.

Editorial note. The five references are two to results of other chapters on categories of finitely presented modules and one on exactness of filtered colimits, plus two to earlier results of this chapter, among them the Hom complex computation that is itself the subject of a card of this lane. All five are navigation stubs with their roles recorded in the reasons.

Editorial note. The preamble span runs to line 90, the current end of the retained preamble, which includes the appended environment definitions; the counted text does not use them, and the span is taken whole so that the file version is unambiguous.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key and by this note. The counted statement and proof cite no bibliography entry, so this card needs no bibliography.

Allowed standard prerequisites (editorial).

- The category of finitely presented modules over a filtered colimit of rings is the colimit of the categories of finitely presented modules over the stages. invoked at proof lines 22129-22132. Reduces the description of finitely presented modules to the stagewise description.
- The category of flat finitely presented modules is likewise the colimit of the corresponding stagewise categories, and relative perfectness is characterized by a structure statement for such modules. used at proof lines 22133-22140. Produces part (1) by taking a representative of the relatively perfect object at some stage.
- The Hom complex between a pseudo-coherent complex and a complex of finite tor dimension is a bounded below complex with flat terms whose cohomology computes the Hom groups, and tensor products commute with filtered colimits which are exact. used at proof lines 22142-22160. Computes the Hom groups over the colimit as the colimit of the Hom groups over the stages, which is part (2).

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 05N7: sources/stacks-chow/tags.txt, source lines 7296–7296. The reference identifies the category of finitely presented modules over the colimit with the colimit of the stagewise categories, invoked at proof lines 22129-22132; the stagewise descent built on it is inside the counted span.

Source reference. Original lemma 02JO: sources/stacks-chow/tags.txt, source lines 3343–3343. The reference supplies the corresponding description for flat finitely presented modules, used at proof lines 22133-22135; part (1) is derived from it inside the counted span.

Source reference. Original lemma 0DHV: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21980–21980. The reference is the structure characterization applied at proof lines 22136-22139 to descend the relatively perfect object to a stage; it is a same-chapter pointer and part (1) is completed inside the counted span.

Source reference. Original lemma 0E1U: sources/stacks-more-algebra/pinned-original-source.tex, source lines 22040–22040. The reference provides the representatives and the Hom complex whose cohomology computes the Hom groups, used at proof lines 22142-22156; the colimit computation built on it is inside the counted span. The lemma itself is the subject of a separate card of this lane with disjoint coordinates.

Source reference. Original lemma 00DB: sources/stacks-chow/tags.txt, source lines 491–491. The reference gives exactness of filtered colimits, used at proof lines 22158-22160 together with the commutation of tensor products with colimits to identify the group over the colimit; it is the last step and the rest is inside the counted span.

Lemma 0.2. *Let $R = \operatorname{colim}_{i\in I} R_i$ be a filtered colimit of rings. Let $0 \in I$ and $R_0 \to A_0$ be a flat ring map of finite presentation. For $i \geq 0$ set $A_i = R_i \otimes_{R_0} A_0$ and set $A = R \otimes_{R_0} A_0$.*

1. *Given an R-perfect K in $D(A)$ there exists an $i \in I$ and an R_i-perfect K_i in $D(A_i)$ such that $K \cong K_i \otimes^{\mathbf{L}}_{A_i} A$ in $D(A)$.*
2. *Given $K_0, L_0 \in D(A_0)$ with K_0 pseudo-coherent and L_0 finite tor dimension over R_0, then we have*

$$\operatorname{Hom}_{D(A)}(K_0 \otimes^{\mathbf{L}}_{A_0} A, L_0 \otimes^{\mathbf{L}}_{A_0} A) = \operatorname{colim}_{i\geq 0} \operatorname{Hom}_{D(A_i)}(K_0 \otimes^{\mathbf{L}}_{A_0} A_i, L_0 \otimes^{\mathbf{L}}_{A_0} A_i)$$

In particular, the triangulated category of R-perfect complexes over A is the colimit of the triangulated categories of R_i-perfect complexes over A_i.

Proof. By Algebra, Lemma 05N7 the category of finitely presented A-modules is the colimit of the categories of finitely presented A_i-modules. Given this, Algebra, Lemma 02JO tells us that category of R-flat, finitely presented A-modules is the colimit of the categories of R_i-flat, finitely presented A_i-modules. Thus the characterization in Lemma 0DHV proves that (1) is true.

To prove (2) we choose $P_0^\bullet$ representing K_0 and $F_0^\bullet$ representing L_0 as in Lemma 0E1U. Then $E_0^\bullet = \operatorname{Hom}^\bullet(P_0^\bullet, F_0^\bullet)$ satisfies

$$H^0(E_0^\bullet \otimes_{R_0} R_i) = \operatorname{Hom}_{D(A_i)}(K_0 \otimes^{\mathbf{L}}_{A_0} A_i, L_0 \otimes^{\mathbf{L}}_{A_0} A_i)$$

and

$$H^0(E_0^\bullet \otimes_{R_0} R) = \operatorname{Hom}_{D(A)}(K_0 \otimes^{\mathbf{L}}_{A_0} A, L_0 \otimes^{\mathbf{L}}_{A_0} A)$$

by the lemma. Thus the result because tensor product commutes with colimits and filtered colimits are exact (Algebra, Lemma 00DB). □